

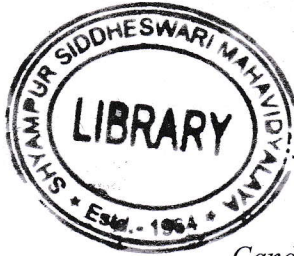
2022

COMPUTER SCIENCE — GENERAL

Paper : DSE-A-3

(Computer Graphics)

Full Marks : 50



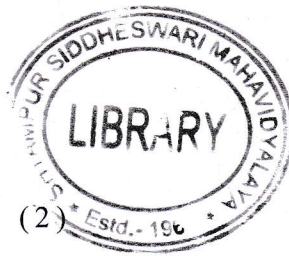
The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any four** from the rest.

1. Answer **any five** questions : 2×5
 - (a) What is image resolution?
 - (b) What is shearing?
 - (c) What are the names of different color models?
 - (d) What is the importance of look-up table in color mapping?
 - (e) Write the properties of monochrome monitor.
 - (f) Define raster scan.
 - (g) What are the advantages of LCD?
 - (h) Differentiate between random scan display and vector scan display.
2. (a) Write the working principle of shadow-masking method.
(b) Briefly describe raster scan display method. 5+5
3. (a) Write the mid-point circle drawing algorithm.
(b) Using midpoint circle drawing algorithm plot a circle whose radius = 10 units. 5+5
4. (a) Discuss Cohen-Sutherland line clipping algorithm.
(b) Apply Cohen-Sutherland line clipping algorithm to clip the line segment with co-ordinates (30, 40) and (60, 30) against the window (20, 20), (60, 20), (60, 40) and (20, 40). 5+5
5. (a) Write a short note on CRT monitor.
(b) Define aspect ratio of a monitor.
(c) Compare advantages and limitations of DDA and Bresenham's line drawing algorithm. 4+2+4

Please Turn Over



X(5th Sm.)-Computer Sc.-G/DSE-A-3/CBCS

6. (a) Why are homogeneous co-ordinates used for transformation computations in computer graphics?
(b) Translate the square ABCD whose co-ordinates are A(0, 0), B(3, 0), C(3, 3) and D(0, 3) by 2-units in both directions and then scale it by 1.5 units in x-direction and 0.5 units in y-direction.
(c) What is inverse transformation? 3+5+2
7. (a) Prove that 2-dimensional rotation and scaling computative if $s_x = s_y$ or $\theta = n\pi$, $n = 0, 1, 2, \dots$
(b) Write down the steps for reflection about an arbitrary line $y = mx + c$. 5+5
8. (a) What do you mean by projection?
(b) Differentiate between parallel projection and perspective projection.
(c) Write short notes on view plane and window. 2+4+4
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